**Theorem:-**

Let $T : V \longrightarrow U$ be a linear transformation and $r(T) = r$. Then $\exists$ basis of $V$ and $U$ such that the matrix representation of $T$ has the following form.

$$A = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

$\dim(V) = m$, $\dim(U) = n$.

**Sol?-**

$$n(T) = m - r$$

Suppose $\{v_{m-r+1}, v_{m-r+2}, \ldots, v_m\}$ is the basis of $\ker(T)$

$$T(v_i) = 0 \quad , \quad i = m-r+1, m-r+2, \ldots, m$$

Extend the basis of $\ker(T)$ as

$$\{v_{m-r+1}, \ldots, v_m, v_1, v_2, \ldots, v_{m-r}\}$$

$$T(v_{m-r+1}) = u_1$$

Let $\{u_1, u_2, \ldots, u_r\}$ is the basis of $R(T)$

Let's extend the basis of $R(T)$ to basis of $U$ as

$$\{u_1, u_2, \ldots, u_r, u_{r+1}, \ldots, u_n\}$$

$$T(v_{m-r+1}) = u_1 = 1 \cdot u_1 + 0 \cdot u_2 + \ldots + 0 \cdot u_n$$

$$T(v_{m-r}) = u_r = 0 \cdot u_1 + \ldots + 1 \cdot u_r + \ldots + 0 \cdot u_n$$

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